Communication and collaboration improve between teams, which improves existing culture.

- New applications in recent times include a lot of components such as middleware, mobile apps, and content management systems. These applications need high availability and fault tolerance. The cloud can help to make the infrastructure or platform resources available within no time.
- The self-service portal or approach available in the cloud inspires teams to implement DevOps practices on their own.
- Cloud automation or Infrastructure as a Code should become a part of the culture of an organisation to eliminate manual activities in installation and configuration.
- DevOps practices also include measurement and monitoring. Both include multiple aspects such as application and cloud resources. Measurement and monitoring not only help in achieving effective outcomes, but also help to troubleshoot issues or to implement preventive measures.
- Cultural transformation is all about identifying the bottlenecks in the various phases of application life cycle management, and finding the solutions to remove these.
  - Resource acquisition and provisioning, along with installation and configuration, take up a lot of time in traditional environments – cloud computing is the answer to this.
  - Cloud service models and cloud deployment models can be used by an organisation to remove all bottlenecks, with proper security and governance.
- Multiple cloud service providers offer DevOps services; so application life cycle phases and cloud resources can be managed effectively from the same paradigm. Here is a quick look at the features of an Azure DevOps portal.
  - *Azure Boards:* These manage continuous planning – Agile, Scrum, CMMI, and so on.
  - *Azure Repos:* These comprise the distributed version control system, centralised version control system in the form of Git, and team foundation version control system.
  - *Azure Pipelines:* These provide build and release definitions to automate the integration and delivery process of an application.
  - *Azure Test Plans:* These include test plans, and test case execution and recording.
  - *Azure Artifacts:* This feature manages Project Artifacts effectively in the Azure DevOps portal.
- Most of the automation servers such as Jenkins, TeamCity, Atlassian Bamboo, Azure DevOps and so on provide ways to deploy services in the cloud environment, or offer integration with cloud environments to launch build or release agents.
- The implementation of DevOps practices makes project teams realise why cloud resources or containers are more useful in the cultural or digital transformation process. If the runtime environment or platform is not available in a timely manner, then all the automation processes are halted.
- True digital transformation is all about finding bottlenecks, finding solutions, creating a roadmap and implementing an automated approach, with processes in place to gain efficiency, quality and faster time to market.

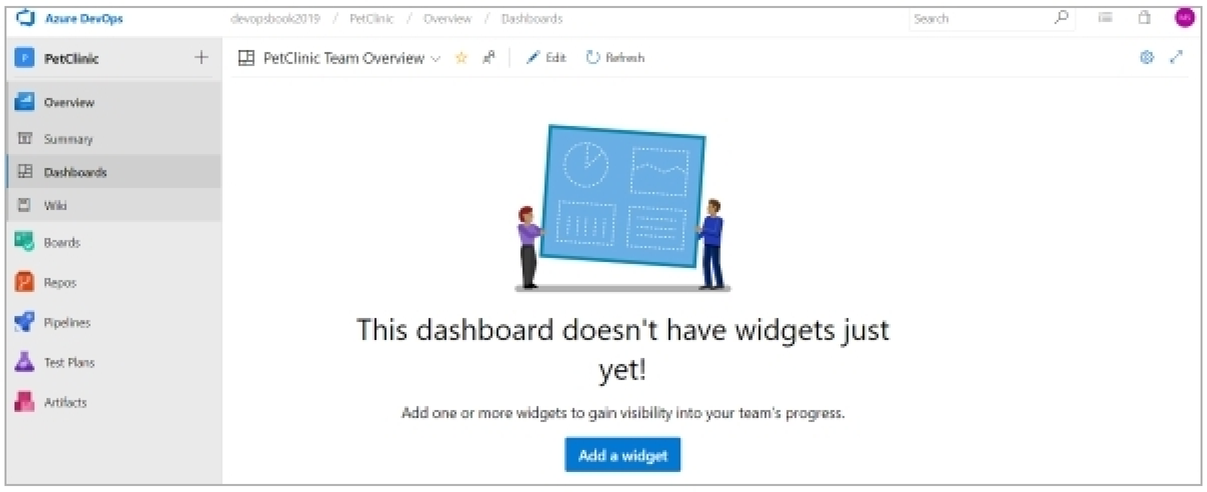

Figure 5: Azure DevOps

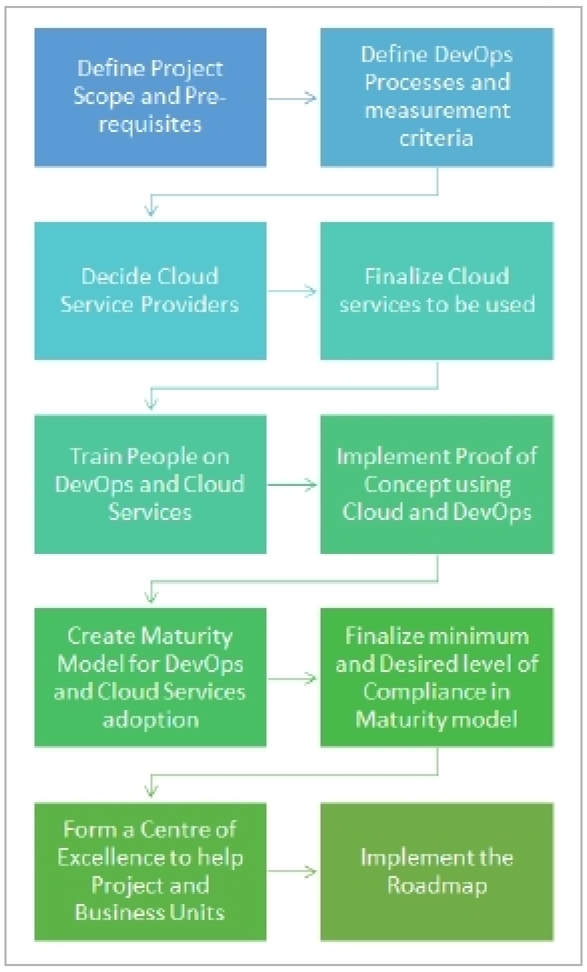

Figure 6: DevOps and cloud adoption

The implementation of DevOps practices, along with faster resource provisioning from the cloud, helps in the digital transformation of an organisation like never before. Additionally, cloud service providers provide hundreds of different services that can be utilised in application architecture to make things more efficient. It is pretty obvious that DevOps and cloud computing complement each other and lead to productivity gains, faster time to market and better quality products. END

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